INCH-POUND
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DETAIL SPECIFICATION

CLOTH, OXFORD, COTTON WARP AND NYLON FILLING, WATER REPELLENT TREATED

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers oxford cloth made of cotton warp and nylon filling yarn, treated with a water repellent treatment.

1.2 Classification. The cloth is available in the following classes as specified (see 6.2).

1.2.1 Classes.

Class 1 - Bleached white
Class 2 - Dyed
Class 3 - Woodland Camouflage Pattern
Class 4 - Desert Camouflage Pattern (6-color)
Class 5 - Desert Camouflage Pattern (3-color)
Class 6 - Operational Camouflage Pattern (OCP)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A- 50199 - Thread, Polyester Core, Cotton or Polyester-Covered

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND SOLDIER CENTER

2-1-1516B	- Woodland Camouflage Pattern (60 inches)
2-1-1884	- Desert Camouflage Pattern (6-color)
2-1-2240	- Desert Camouflage Pattern (3-color)
2-1-2592	- Operational Camouflage Pattern (OCP)

(Comments, suggestions, or questions on this document should be addressed to: Department of the U.S. Army, DEVCOM - Soldier Center, ATTN: FCDD-SCD-PE, 10 General Greene Avenue, Natick, MA 01760-5019.)

FEDERAL TRADE COMMISSION

16 CFR Part 303 Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Evaluation Procedure for Gray Scale for Color Change
AATCC EP2 - Evaluation Procedure for Gray Scale for Staining
AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM8 - Test Method for Colorfastness to Crocking: Crockmeter Method
AATCC TM15 - Test Method for Colorfastness to Perspiration
AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon Arc
AATCC TM20 - Test Method for Fiber Analysis: Qualitative
AATCC TM22 - Test Method for Water Repellency: Spray
AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated
AATCC TM70 - Test Method for Water Repellency: Tumble Jar Dynamic Absorption
AATCC TM81 - Test Method for pH of the Water-Extract from Wet Processed Textiles
AATCC TM118 - Test Method for Oil Repellency: Hydrocarbon Resistance
AATCC TM127 - Test Method for Water Resistance: Hydrostatic Pressure
AATCC TM135 - Test Method for Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D737 - Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1683/D1683M - Standard Test Method for Failure in Sewn Seams of Woven Fabrics
ASTM D3775 - Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M - Standard Test Methods for Mass per Unit Area (Weight) of Fabric
ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of these documents are available online at https://www.astm.org.)

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OTHER PUBLICATIONS

Fabric Defect Replica Scales

(For Information on obtaining copies of the Fabric Defect Replica Scales please contact U.S.Army, DEVCOM – Soldier Center, ATTN: FCDD-SCP-PE, 10 General Greene Avenue, Natick, MA 01760-5019.)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies of this document are available online at https://www.crcpress.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. Unless otherwise indicated, the finished cloth shall match the standard sample for shade and appearance, and shall, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Cotton warp yarn. The cotton shall be carded, combed, drawn, spun and twisted into two-ply yarns.

3.4.2 Nylon filling yarn. The nylon shall be semi-dull continuous multi-filament.

3.4.3 Weave. The weave shall be oxford (2-warp ends weaving as one (1)).

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3.5 Color.

3.5.1 Color, Class 1. The color of Class 1 cloth shall be bleached white to match the standard sample (see 6.3) and shall be produced by chemical bleaching using a peroxide-type agent and a fluorescence optical brightener with a resulting peak emission in the blue violet. The hue of the fluorescence shall be the same as that of the standard sample when tested as specified in 4.5.

3.5.2 Color, Class 2. The cloth shall be dyed and shall match the standard test when tested as specified in 4.5.

3.5.3 Color, Class 3 (Woodland Camouflage Pattern print). The finished cloth shall be either dyed to a ground shade matching Light Green 354 and then overprinted with the remaining three (3) colors for the Dark Green 355, Brown 356, and Black 357 areas of the camouflage pattern (see Drawing 2-1-1516B) or when the ground shade is not dyed to match Light Green 354, all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors of the pattern. Resin bonded pigments are not permitted.

3.5.4 Color, Class 4 (Desert Camouflage Pattern, 6-color). The finished cloth shall be either dyed to a ground shade matching light Tan 379 and then overprinted with the remaining five (5) colors for the Tan 380, Light brown 381, Dark Brown 382, Black 383 and Khaki 384 areas of the camouflage pattern (see Drawing 2-1884) or when the ground shade is not dyed to match Light Tan 379, then all six (6) colors of the camouflage pattern shall be printed to match all the six (6) colors. Resin bonded pigments are not permitted.

3.5.5 Color, Class 5 (Desert Camouflage Pattern, 3-color). The finished cloth shall be either dyed to a ground shade matching Light Tan 492 and then overprinted with the remaining two (2) colors for the Light Brown 493 and Light Khaki 494 areas of the camouflage pattern (see Drawing 2-1-2240) or when the ground shade is not dyed to match Light Tan 492, then all three (3) colors of the camouflage pattern shall be printed to match all three (3) colors of the pattern. Resin bonded pigments are not permitted.

3.5.6 Color, Class 6 (Operational Camouflage Pattern, (OCP)). The finished cloth shall be either dyed to a ground shade matching Dark Cream 559 and then overprinted with the remaining six (6) colors for the Tan 525, Light Sage 560, Olive 527, Dark Green 528, Brown 529 and Bark Brown 561 areas of the camouflage pattern (see Drawing 2-1-2592) or when the ground shade is not dyed to Dark Cream 559 then all seven (7) colors of the camouflage pattern shall be printed to match all seven (7) colors of the pattern. Resin bonded pigments shall not be used.

3.5.7 Visual shade matching. The color and appearance of the dyed and printed cloth shall match the standard sample when tested as specified in 4.5.

3.5.8 Labile sulfur. Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used. The dyed cloth shall be “Free” of labile sulfur when tested as specified in 4.5.

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3.5.9 Colorfastness. Unless otherwise specified in the contract or procurement documents, the finished dyed and printed cloth shall conform to the colorfastness requirements listed below in Table I when tested as specified in 4.5.

TABLE I. Colorfastness requirements.

Class	Colors evaluation	Laundering (Color Change and Staining) (3 cycles) (min.)	Light (after 40 AFU or 170 kJ/(m^2nm)@420 nm) (min) 1/	Perspiration (Acid & Alkaline) (Color Change and Staining) (min.)	Crocking (Dry/Wet) (min.)
Class1	-	3-4	3-4	3-4	
Class 2	All colors	3-4	3-4	3-4	3.5
Class3	All colors except black	3-4	3-4	3-4	3.5
	Black	3-4	3-4	3-4	1.5
Class 4	All colors except black	3-4	3-4	3-4	3.5
	Black	3-4	3-4	3-4	1.5
Class 5 and 6	All colors	3-4	3-4	3-4	3.5

1/ AFU: AATCC Fading Units

3.5.10 Pattern execution. The pattern on the printed finished cloth(s) shall match the standard sample with respect to design, colors, and registration of the respective areas. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations and/or feathering as required for each color. Each pattern area shall show full coverage; skitteriness, feathering, haloing or trapping and off register exceeding that shown on the standard sample in any of the printed areas will not be acceptable. When the standard sample is not available for pattern execution, a pattern drawing will be provided (see 6.3) and the pattern on the finished cloth shall match that of the drawing(s) (see 2.2.2 and 6.4) when tested as specified in 4.5. The pattern repeat shall be as follows:

Woodland Camouflage Pattern - 27.25 (+1.25, -2.50) inches in the warp direction.
Desert Camouflage Pattern (3 color) - 16.75 (+1.25, -2.25) inches in the warp direction.
Desert Camouflage Pattern (6-color) - 16.75 (+1.25, -1.75) inches in the warp direction.
Operational Camouflage Pattern (OCP) - 25.25 (+1.25, -2.50) inches in the warp direction and 68-inches maximum camouflage printed area in the filling direction.

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3.5.11 Spectral reflectance.

3.5.11.1 Class 3, Woodland Camouflage Pattern. The spectral reflectance values shall conform to the requirements specified in Table II, when tested as specified in 4.5.

TABLE II. Spectral reflectance requirement (Class 3).

Reflectance values (percent) - Woodland Camouflage Pattern						
Wavelength Nanometers (nm)	Light Green 354		Dark Green 355 and Brown 356		Black 357	
	min.	max.	min.	max.	min.	max.
600	8	18	3	9	-	10
620	8	18	3	9	-	10
640	8	18	3	9	-	10
660	8	18	3	12	-	10
680	10	22	3	14	-	10
700	18	33	5	18	-	10
720	22	45	7	20	-	10
740	30	55	12	28	-	10
760	35	65	18	36	-	10
780	40	75	26	44	-	10
800	45	80	34	52	-	10
820	50	86	42	60	-	10
840	55	88	50	68	-	10
860	60	90	56	74	-	10

3.5.11.2 Class 4, Desert Camouflage Pattern (6-color). The spectral reflectance values, except for the Black 383, shall conform to the requirements specified in Table III, when tested as specified in 4.5.

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TABLE III. Spectral reflectance requirement (Class 4).

Reflectance values (percent) – Desert Camouflage Pattern (6-color)						
Wavelength Nanometers (nm)	Light Tan 379		Tan 380 and Khaki 384		Light Brown 381 and Dark Brown 382	
	min.	max.	min.	max.	min.	max.
700	38	53	25	44	19	41
720	38	54	25	45	20	41
740	39	55	25	46	20	42
760	40	56	26	47	21	42
780	41	57	27	48	21	42
800	43	58	28	50	22	43
820	45	59	30	52	23	45
840	48	62	33	55	24	46
860	50	65	36	58	25	48

3.5.11.3 Class 5, Desert Camouflage Pattern (3-color). The spectral reflectance values shall conform to the requirements specified in Table IV, when tested as specified in 4.5.

TABLE IV. Spectral reflectance requirement (Class 5).

Reflectance values (percent) – Desert Camouflage Pattern (3-color)						
Wavelength, Nanometers (nm)	Light Tan 492		Light Brown 493		Light Khaki 494	
	Min.	Max.	Min.	Max.	Min.	Max
700	38	53	19	41	25	44
720	38	54	20	41	25	45
740	39	55	20	42	25	46
760	40	56	21	42	26	47
780	41	57	21	42	27	48
800	43	58	22	43	28	50
820	45	59	23	45	30	52
840	48	62	24	46	33	55
860	50	65	25	48	36	58

3.5.11.4 Class 6, Operational Camouflage Pattern (OCP). The spectral reflectance values shall conform to the requirements specified in Table V, when tested as specified in 4.5.

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TABLE V. Spectral reflectance requirement (Class 6).

Reflectance values (percent) – Operational Camouflage Pattern (OCP)						
	Dark Cream 559 & Tan 525		Light Sage 560, Olive 527 & Brown 529		Dark Green 528 & Bark Brown 561	
Wavelength, nanometers (nm)	Min,	Max.	Min.	Max.	Min.	Max.
600	22	44	12	30	3	11
620	24	45	12	30	3	12
640	24	45	12	32	4	13
660	25	45	12	32	4	14
680	28	45	14	34	4	17
700	28	48	14	36	6	23
720	30	52	16	39	6	23
740	32	55	18	41	10	25
760	36	56	20	43	14	30
780	38	57	22	45	18	35
800	40	57	22	45	21	40
820	44	58	24	46	24	42
840	46	59	26	47	26	43
860	48	60	28	48	28	45

3.6 Physical requirements. The finished cloth shall conform to the colorfastness and physical requirements listed below in Table VI, when tested in accordance with the test methods specified in 4.5.

TABLE VI. Physical requirements.

Characteristic	Requirement				
	Class1	Class 2	Class 3	Class 4, 5 and 6	
Weight, oz./yd^2					
Min.	4.8	4.8	4.8	4.8	4.8
Max	5.8	5.8	-	-	-
Yarns per inch (min.)					
Warp	160	160	160	160	160
Filling	74	74	74	74	74

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TABLE VI. Physical requirements. - Continued

Characteristic	Requirement				
	Class1	Class 2	Class 3	Class 4, 5 and 6	
Breaking strength, pounds, (min.)					
Warp direction	140	150	150	150	150
Filling direction	125	125	125	125	125
Dimensional change (percent) (max)					
Warp	2.0	2.0	2.0	2.0	2.0
Filling	2.0	2.0	2.0	2.0	2.0
Air permeability, $ft^3/min/ft^2$ (max)					
After 3 launderings	3.0	3.0	4.0	4.0	4.0
pH (range)	5.0-8.5	5.0-8.5	5.0-8.5	5.0-8.5	5.0-8.5
Seam efficiency (percent) (min.)	70	70	70	70	70

3.7 Finish. The cloth shall be given a durable fluorocarbon type water repellent treatment. The finished cloth shall meet the requirements specified in Table VII when tested as specified in 4.5.

TABLE VII. Water repellency.

	Spray Rating 1/	Hydrostatic height, (cm)		Dynamic absorption (percent)		Resistance to organic liquid
		Min. lot average	Min. 2/	Max lot average	Max 3/	Min.
Initial	90,90,80	35	30	20	25	No wetting to n-tetradecane
After 3 launderings	-	35	30	-	-	-
After 15 launderings	-	-	-	20	25	No wetting to n-tetradecane

1/ Three (3) individual determinations shall be equal to or better than the ratings specified.
2/ No individual specimen shall fall below the specified minimum.
3/ No individual specimen shall exceed the specified maximum.

3.8 Marking. The face side shall be identified by applying a stamping on the face side of the cloth with the word “FACE” at each end of the roll.

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3.9 Width. For Government procurements only, the width of the finished cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage.

3.10 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put-up full width on a roll as specified in contract or purchase order.

3.11 Fiber identification. Each roll of the finished cloth shall be labeled or ticketed for fiber content in accordance with the 16 CFR Part 303 Rules and Regulations Under the Textile Fiber Products Identification Act.

3.12 Toxicity. The finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.7. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used (see 6.6).

3.13 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects as specified in 4.3 and 4.4, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual examination. Each roll in the sample shall be examined yard by yard on the face side for defects in accordance with MIL-STD-3064, Type I.

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4.4.1 Roll identification and marking examination. During the yard-by-yard examination each roll in the sample shall be examined for roll identification and marking defects as specified in MIL-STD-3064.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination (individual and total yardage). During the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 Material testing. The cloth shall be tested for the characteristics listed in Table VIII. The methods of testing as specified wherever applicable and as listed in Table VIII shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE VIII. Material tests.

Characteristic	Requirement Reference	Test Method
Fiber identification Cotton fiber Nylon fiber	See 3.4.1 3.4.2	 AATCC TM20 (see 6.4) AATCC TM20 (see 6.4)
Weave	See 3.4.3	Visual 1/
Fluorescence (Class 1)	See 3.5.1	See 4.6.1.1
Visual shade matching	See 3.5.7	See 4.6.1
Labile sulfur	See 3.5.8	See 4.6.2
Colorfastness: Laundering Light Perspiration Crocking	Table I	 AATCC TM61, Test 2A 2/ AATCC TM16.3, Opt. 3 3/ AATCC TM15 2/ AATCC TM8 4/
Pattern execution	See 3.5.10	Visual

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TABLE VIII. Material tests. - Continued

Characteristic	Requirement Reference	Test Method
Spectral reflectance (Class 3, 4, 5, and 6)	Table II, Table III, Table IV and Table V	See 4.6.3
Weight	Table VI	ASTM D3776/D3776M (Opt. C)
Yarns per inch	Table VI	ASTM D3775
Breaking strength	Table VI	ASTM D5034
Dimensional change	Table VI	AATCC TM135 (3)(V)(Aiii)
Air permeability	Table VI	ASTM D737
pH	Table VI	AATCC TM81
Seam efficiency	Table VI	ASTM D1683/D1683M 5/
Spray rating	Table VII	AATCC TM22
Hydrostatic height Initial After 3 launderings	Table VII	 AATCC TM127 AATCC TM 135 (3)(V)(Aiii) and AATCC TM 127
Dynamic absorption Initial After 15 launderings	Table VII	 AATCC TM70 AATCC TM135 (3)(V)(Aiii) and AATCC TM70
Resistance to organic liquids Initial After 15 launderings	Table VII	 AATCC TM118 AATCC TM135 (3)(V)(Aiii) and AATCC TM 118
Toxicity	See 3.12	See 4.7

1/ One (1) determination shall be made from each sample and results reported as “pass” or fail”
2/ Rated using the AATCC EP1, and AATCC EP2
3/ Rated using the AATCC EP1
4/ Rated using the AATCC EP8
5/ The needle shall measure 0.040 inch ± 0.001 inch across blade at the eye. The thread shall be polyester-covered, or cotton-covered in accordance with A-A-50199, Tex 60 for the needle and Tex 45 for the looper.

4.6 Methods of testing and inspection.

4.6.1 Visual shade matching. The color and appearance of the material shall match the standard sample when viewed using the AATCC EP9 Option C (see 6.5), with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 (± 200)K.

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4.6.1.1 Determination of fluorescence (Class 1 only). A specimen of the sample unit and a specimen of the standard sample shall be compared under ultra-violet light in an otherwise completely dark room. The specimen shall be considered satisfactory if its hue of fluorescence is the same as the standard sample. One determination shall be performed on the sample unit and the results reported as "pass" or fail".

4.6.2 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin(II) chloride (stannous chloride) solution by dissolving 100 g $SnCl_2$ (ACS grade) in 100 mL of 35% hydrochloric acid (ACS grade) and adding the mixture to 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0 g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The samples shall be submersed the acidic tin(II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin(II) chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as "pass" or "fail.

4.6.3 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometer (nm) for Woodland Camouflage Pattern and the Operational Camouflage Pattern (OCP) and 700 to 860 nanometers (nm) for Desert Camouflage Pattern (3 color and 6-color) at 20 nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white or vitrolite tiles. The spectral band width shall be less than 20 nm at 860 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source D65. The measured areas should be at least 6-inches away from the selvage. The specimen shall be measured as a single layer backed with layers of the same fabric and shade as follows:

The Woodland Camouflage Pattern and the Operational Camouflage Pattern colors shall be measured as a single layer backed with three (3) layers of the same fabric and shade. The Desert Camouflage Pattern (3-color and 6-color) colors shall be measured as a single layer backed with four (4) layers of the same fabric and shade. The spectral reflectance shall be measured on the mostly solid color area not the tonal area. The specimen shall be viewed at an angle no greater than 10 degrees from normal, with the specular component included. Measurements shall be taken on a minimum of two (2) different areas. Specimens shall be oriented in different

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directions during testing. When possible, the specimens tested shall not contain the same warp or filling yarns when presented to the sample port. Photometric accuracy of the spectrophotometer shall be within 1-percent and wavelength accuracy within 2-nanometers. The diameter for standard aperture size used in the color measurement device shall be the largest possible for each color. Any color having spectral reflectance values falling outside the limits at four (4) or more of the wavelengths specified shall be considered a test failure.

4.7 Toxicity test. When required, (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the finish is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If the toxicity requirement (see 3.12) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth is intended for use in the manufacturing of snow mitten shells, field pack cover, hood components, caps and the casualty evacuation bag casing.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Class required (1.2)
c. The specific issue of individual documents referenced (see 2.2)
d. When first article is required (see 3.1)
e. Width of cloth required (see 3.9)
f. Length required if other than specified (see 3.10)
g. When toxicity is required (see 3.12)
h. Conformance inspection (see 4.3)
i. Inspection conditions (see 4.3.1)
j. Packaging (see 5.1)

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6.3 Standard sample. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Visual shade matching. In 2017, Option A of AATCC Evaluation Procedure 9, Option C. NOTE: In case of confusion, the viewing geometry should be such that the specimen plane and illumination source are parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens.

6.6 EPA information is available online at https://www.epa.gov.

6.7 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.8 Subject term (key word) listing.

Camouflage
Field pack cover
Printed cloth
Snow mitten shells

Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing Activity:
DLA-CT

(Project 8305-2021-012)

Review activities:
Army - MD

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.

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